

AI SQL Agent

AI-Powered Natural Language SQL & MongoDB Assistant

Project Overview

AI SQL Agent is an AI-powered database assistant that converts natural language into executable MySQL and MongoDB queries. It integrates multiple AI providers, validates generated queries, automatically repairs common query errors, executes queries safely, and presents results with explanations, analytics, visualizations, and export options.

Objectives

- Convert natural language into SQL and MongoDB queries.
- Support multiple AI providers (Ollama, OpenAI, Anthropic).
- Validate and safely execute generated queries.
- Automatically repair invalid queries.
- Export results to CSV, Excel, and PDF.

Technology Stack

Python, Streamlit, MySQL, MongoDB, Pandas, SQLParse, OpenPyXL, ReportLab, Ollama, OpenAI, Anthropic.

System Architecture

User → Streamlit UI → Agent Orchestrator → SQL/Mongo Agents → Validators → Query Service → MySQL/MongoDB → Explanation Agent → Dashboard & Export.

Core Features

- Natural Language to SQL
- Natural Language to MongoDB
- SQL & MongoDB Validators
- SQL & MongoDB Repair Agents
- Query History & Saved Queries
- Dashboard Analytics
- Interactive Charts
- CSV, Excel & PDF Export

Project Modules

Agents, AI Provider Manager, Database Services, Schema Explorer, Validators, Parsers, Analytics, Utilities, History, Saved Queries, Templates, Suggestions and Streamlit UI.

Future Enhancements

PostgreSQL support, REST API, Docker deployment, User Authentication, Cloud Deployment, Multiple Database Connections and Voice Commands.

Conclusion

AI SQL Agent demonstrates practical integration of Artificial Intelligence with relational and NoSQL databases. The project follows a modular architecture, making it scalable, maintainable, and suitable as a portfolio or academic project.